Beijing Lin

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EDUCATION

University of British Columbia *Sep 2024-May 2029 (Expected)*

Bachelor of Applied Science (B.A.Sc.) in Electrical Engineering — *Co-op*

Relevant courses: Circuit Analysis, Data Structures and Algorithms, Computing Systems I, Electro-Mechanical Energy Conversion and Transmission, Signals and Systems, Engineering Electromagnetics

PROJECTS

Real-Time Tron Light Cycle Game *Nov 2025-Dec 2025*

- Developed a real-time two-player game in C running on an FPGA platform using a memory-mapped VGA graphics interface.
- Implemented hardware timer interrupts and key interrupts to handle game timing, player input, and deterministic movement updates.
- Developed framebuffer-based collision detection and debugged timing, memory and boundary conditions.

FPGA Combination Lock System *Sep 2026-Oct 2026*

- Designed and implemented a digital combination lock in SystemVerilog using an Finite-State Machine (FSM) for password verification and lock-state control.
- Interfaced keypad inputs and seven-segment displays on the DE10-Lite FPGA, implementing input handling, state transitions, and visual status feedback.

EXPERIENCE

UBC Rocket | Avionics Hardware & System Validation *Sep 2025-Apr 2026*

- Contributed to PYRO board schematic design and validation using Altium Designer, component datasheet analysis, and LTspice simulations.
- Configured STM32 peripherals (SPI, GPIO, interrupts) in STM32CubeMX and performed firmware interface testing and debugging using UART and logic-level checks.

UBC Open Robotics | Electrical Engineering Team Member *Jan 2026-Present*

6-DOF Robotic Arm

- Calculated motor and electronic subsystem power requirements for a 6-DOF robotic arm.
- Evaluated battery-to-power-distribution wiring, including wire gauge selection, continuous current capacity, and voltage drop considerations.

Automatic Guitar Tuner

- Designed a PCB for microphone and ESP32 signal routing within mechanical enclosure size constraints
- Integrated power architecture using LiPo battery and DC-DC conversion modules for portable operation

SKILLS

Programming & HDL: C, Java, Python, Verilog/SystemVerilog, RISC-V Assembly
Embedded & Systems: ESP32, STM32, FPGA, Memory-Mapped I/O, SPI/UART, Hardware Interrupts
Circuit Design & EDA: Altium Designer, EasyEDA (JLCEDA), LTspice, Proteus, Quartus, Questa, Circuit Design & Analysis
Hardware & Testing: PCB Assembly, SMD Soldering, Oscilloscope, Multimeter, Signal Generator
Software & Tools: MATLAB/Simulink, Git/GitHub, Linux, Visual Studio Code, Fusion 360, SolidWorks

CERTIFICATIONS & AWARDS

AP Scholar with Distinction (2024) | Governor General's Academic Medal (2023-2024), WHMIS Training